



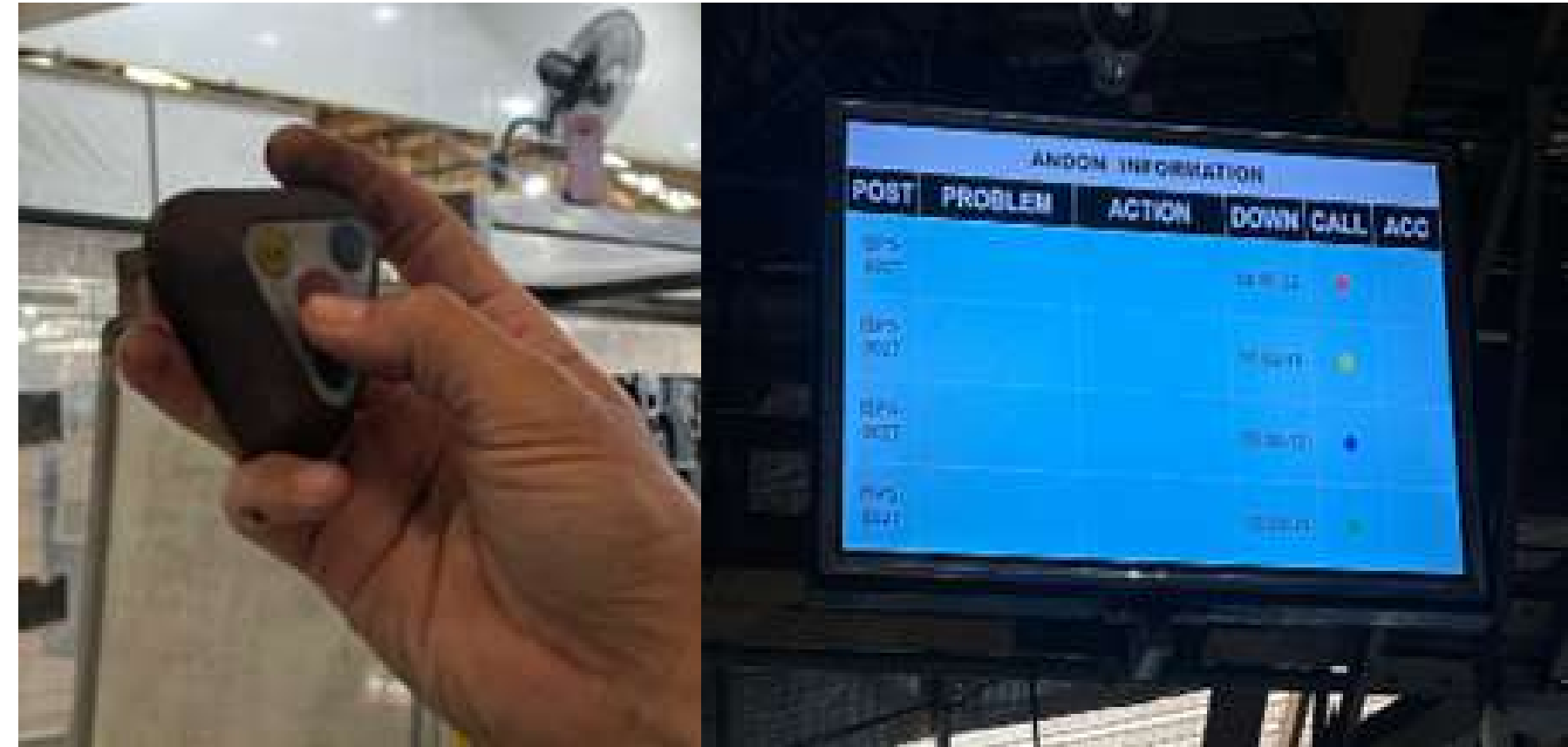
PERANCANGAN IMPLEMENTASI SMART ANDON SYSTEM BERBASIS ESP32 TERINTEGRASI PLC UNTUK MONITORING DOWNTIME MESIN DAN NOTIFIKASI AUDIO-VISUAL BERBASIS IOT

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PT Toyota Motor Manufacturing Indonesia telah menerapkan sistem Andon sebagai media komunikasi antara operator dan tim maintenance ketika terjadi gangguan mesin. Sistem tersebut menggunakan remote Andon yang mampu mengaktifkan alarm sehingga maintenance dapat segera mengetahui adanya downtime mesin.

Meskipun sistem lama telah membantu proses komunikasi, sistem tersebut masih memiliki beberapa keterbatasan, yaitu belum mampu menyimpan histori downtime secara otomatis serta belum menyediakan dashboard monitoring berbasis Internet of Things (IoT). Akibatnya, proses analisis downtime dan evaluasi performa mesin masih dilakukan secara manual.

Untuk mengatasi keterbatasan tersebut, dikembangkan Smart Andon System berbasis ESP32 yang terintegrasi dengan PLC dan platform IoT sehingga data downtime dapat direkam, disimpan, dan dimonitor secara real-time.



REMOTE DAN DASHBOARD LAMA

Existing System vs Smart Andon System



Existing Andon	Smart Andon
Remote Andon	Remote Andon
Alarm Notification	Alarm Notification
Maintenance Call	Maintenance Call
Manual Downtime Record	Automatic Downtime Record
No Historical Database	Historical Database
No IoT Dashboard	IoT Dashboard
Limited Monitoring	Real-Time Monitoring
Difficult Data Analysis	Easy Data Analysis

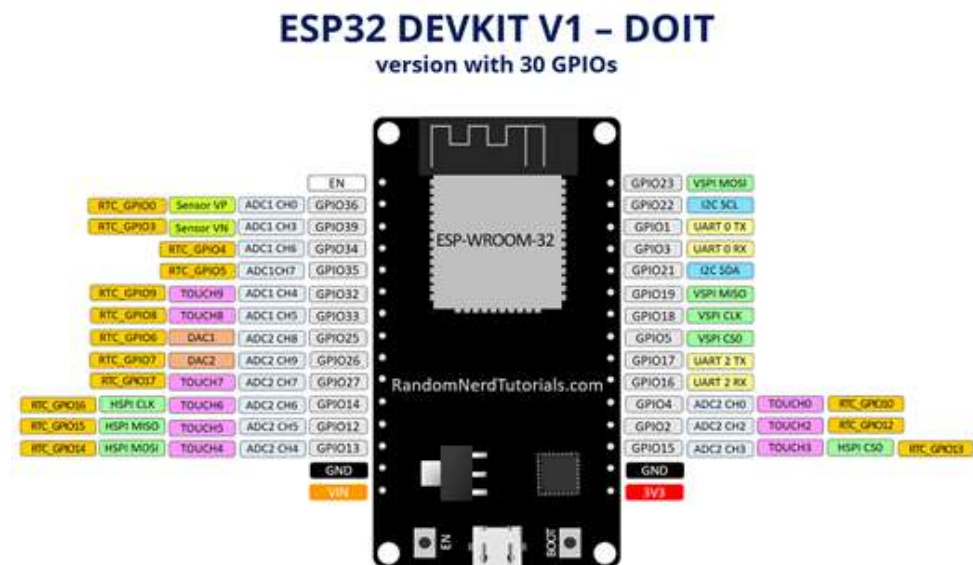
Main Improvement :

- **Update Remote**
- **Alarm tetap digunakan.**
- **Menambahkan penyimpanan data downtime secara otomatis.**
- **Menambahkan dashboard monitoring berbasis IoT.**
- **Menambahkan histori downtime untuk analisis proses.**





ALAT DAN BAHAN



ThingsBoard



MQTT Protocol

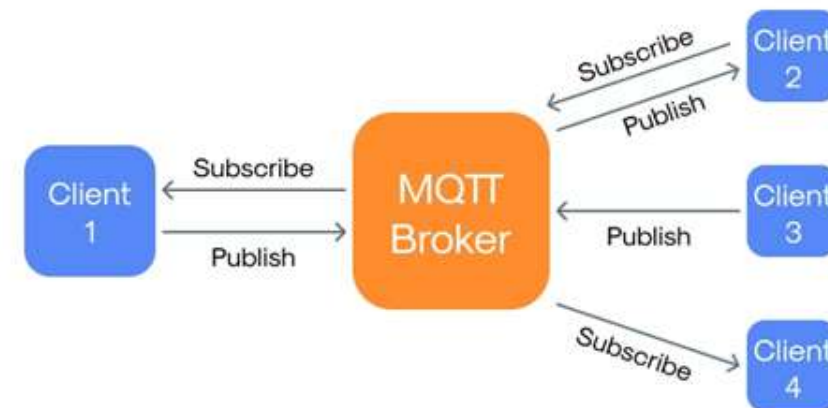




DIAGRAM BLOK SISTEM



SMART ANDON SYSTEM



ALUR KOMUNIKASI

ESP MESIN

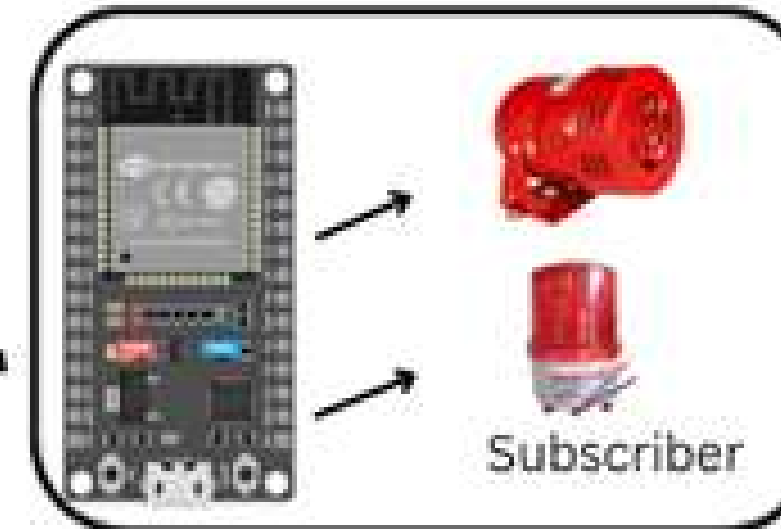
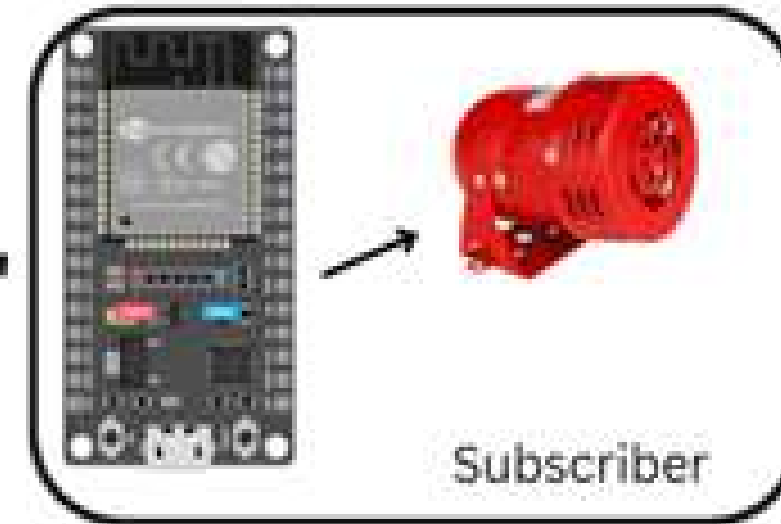


BORKER MQTT



HIVEMQ

ESP LINE



ESP OASIS

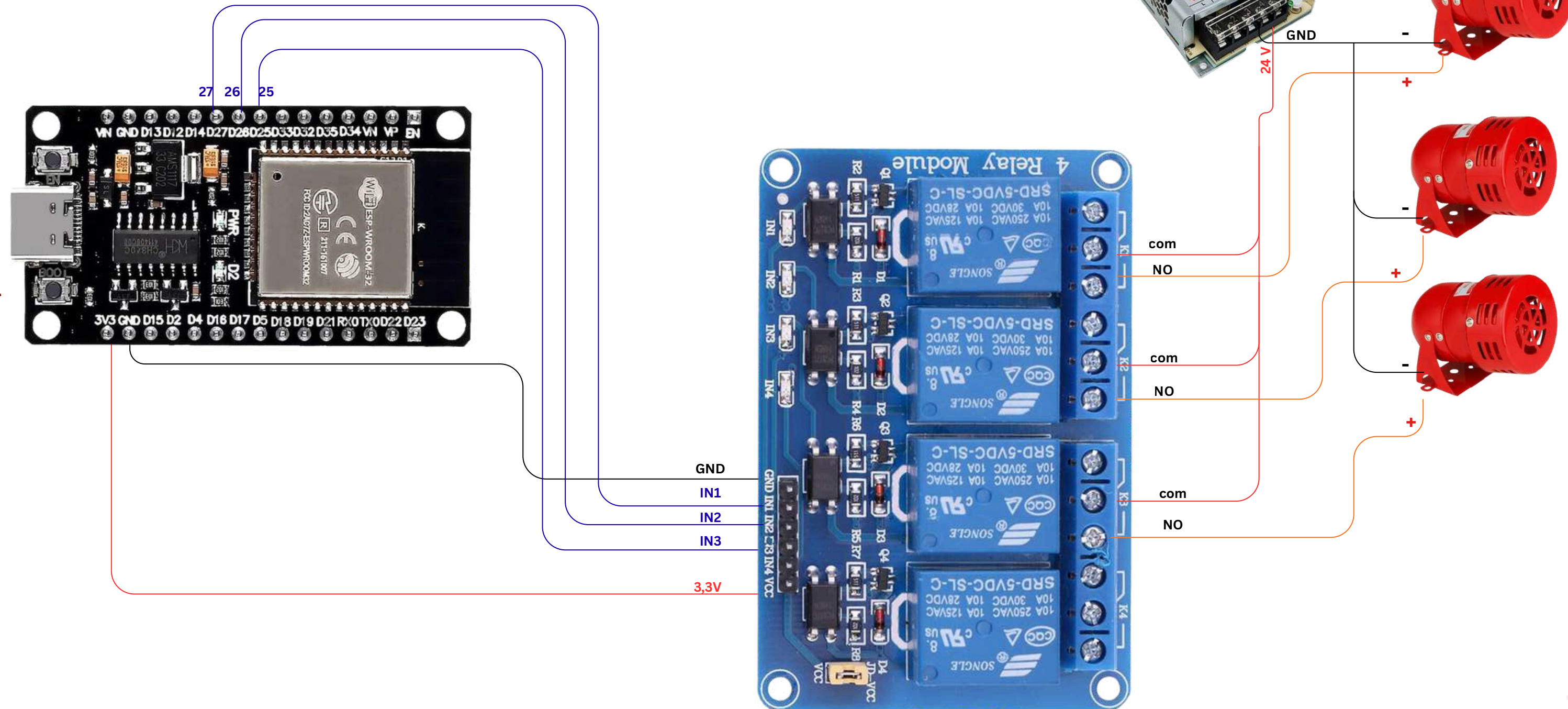




SMART ANDON SYSTEM

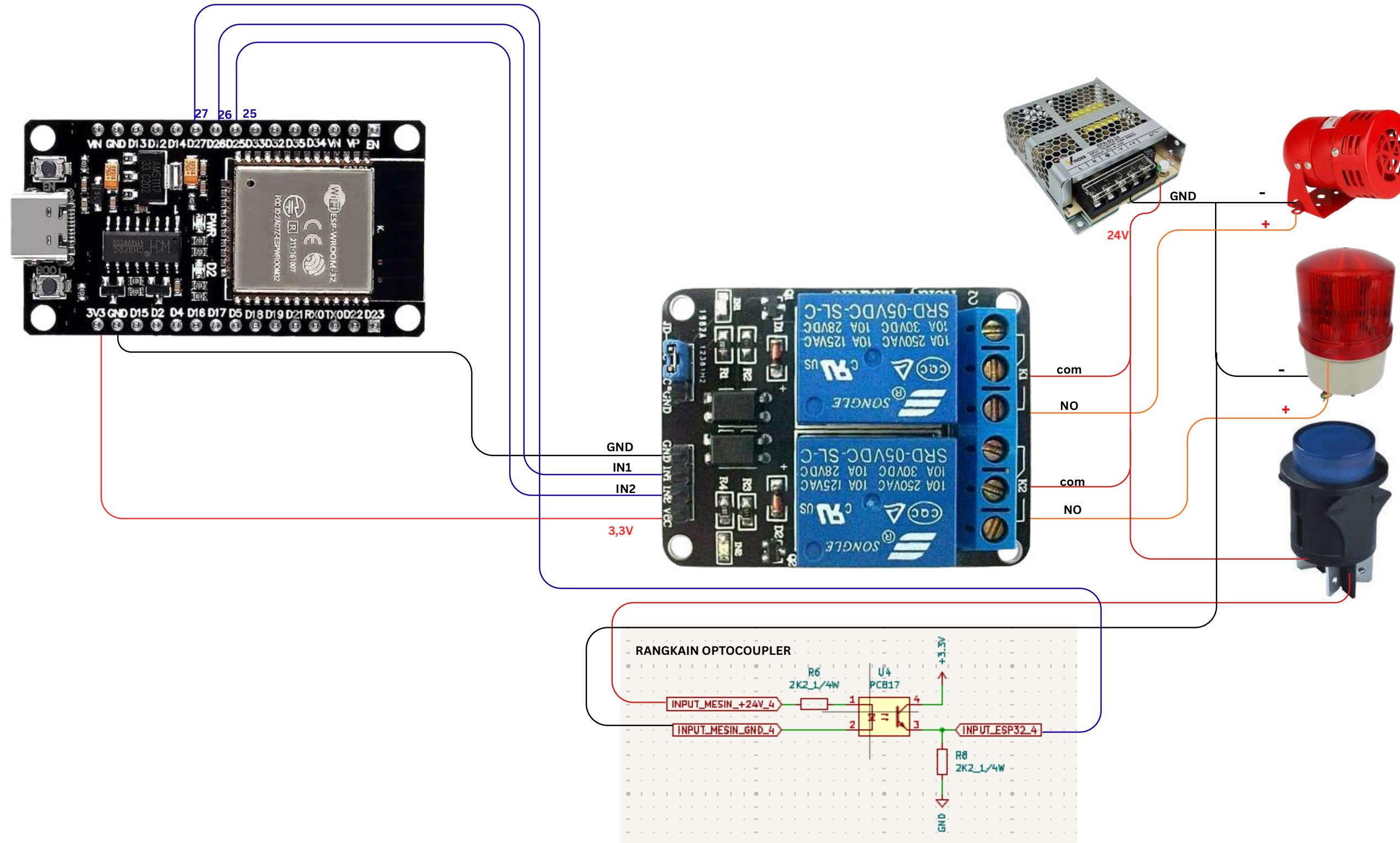


SCHEMATIC ESP LINE





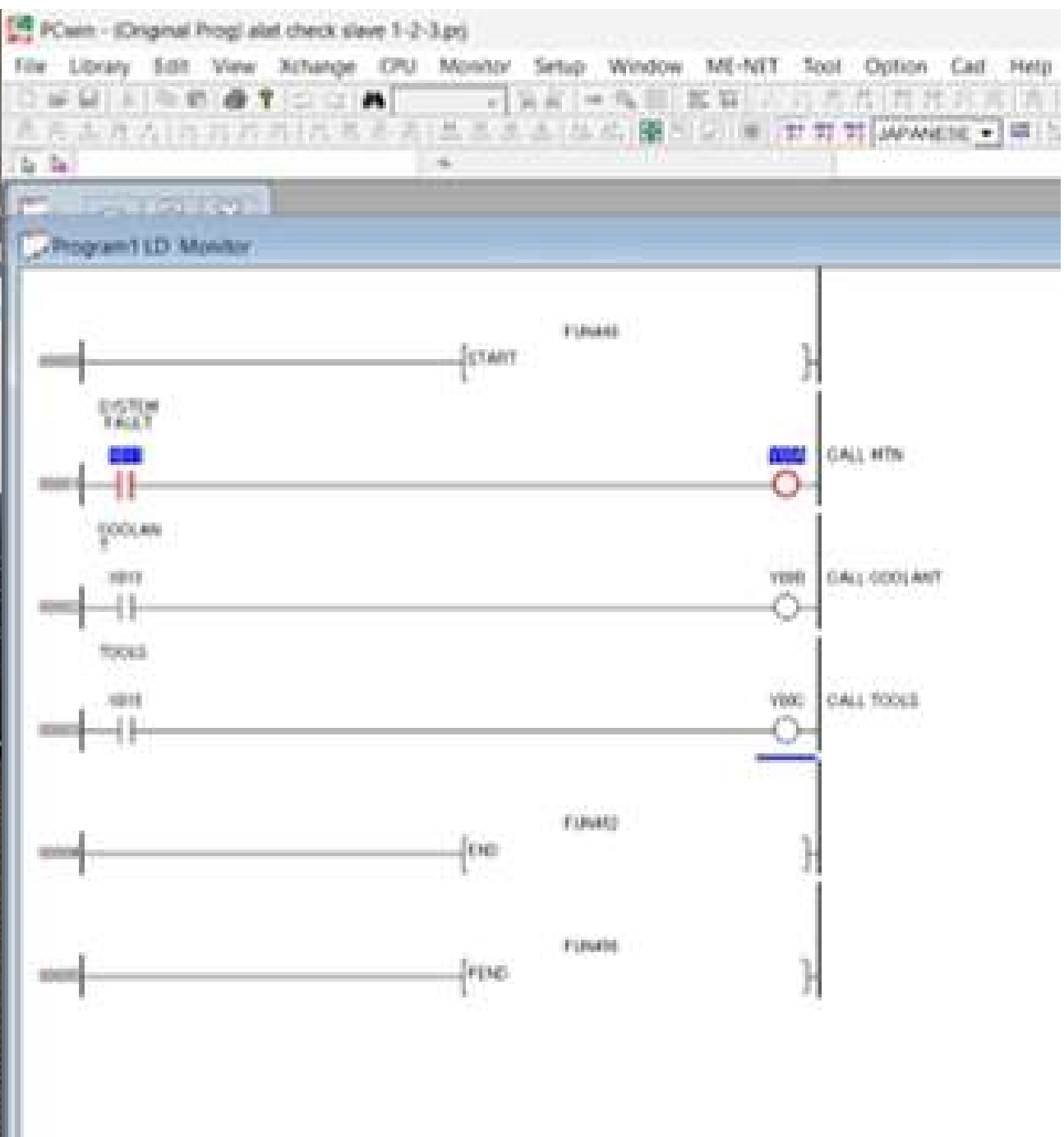
SCHEMATIC ESP OASIS





HASIL PERCOBAAN ESP MESIN PLC









HASIL SERIAL MONITOR ESP MESIN

```
Output Serial Monitor X
Message (Enter to send message to 'ESP32 Dev

CALL : MTN
DURATION AWAL : 00:00:13
DURATION FINISH : 00:00:48

CALL : MTN
DURATION AWAL : 00:00:13
DURATION FINISH : 00:00:49
-----
CALL : MTN
DURATION AWAL : 00:00:13
DURATION FINISH : 00:00:50

CALL : MTN
DURATION AWAL : 00:00:13
DURATION FINISH : 00:00:51
>>> MQTT : FINISH
```

```
Output Serial Monitor X
Message (Enter to send message to 'ESP32 Dev

DURATION FINISH : 00:00:04

CALL : MTN
DURATION AWAL : 00:00:00
DURATION FINISH : 00:00:05
>>> MQTT : MUTE LINE
BUTTON MUTE LINE
```





HASIL SERIAL MONITOR ESP LINE DAN OASIS

```

F (err) {
Serial Monitor X

n to send message to 'ESP32 Dev Module' on 'COM6')

7 -> 192.168.43.185
7 -> Connecting HiveMQ...CONNECTED
7 -> SUBSCRIBE: andon/line1/produksi
5 -> =====
5 -> TOPIC : andon/line1/produksi
5 -> CALL : COOLANT
5 -> STATUS : finish
5 -> BUZZER OFF (FINISH)
5 -> =====
5 -> =====
5 -> TOPIC : andon/line1/produksi
5 -> CALL : PRODUKSI
5 -> STATUS : running
5 -> BUZZER ON (CALL AKTIF)
5 -> =====
  
```

A

```

OASIS BUTTON NYE DITERAK
BUZZER OASIS : OFF
STATUS : NYED
  
```

```

OASIS STATUS : running
OASIS STATUS : finish
OASIS STATUS : running
  
```

```

OASIS BUTTON NYE DITERAK
BUZZER OASIS : OFF
STATUS : NYED
  
```

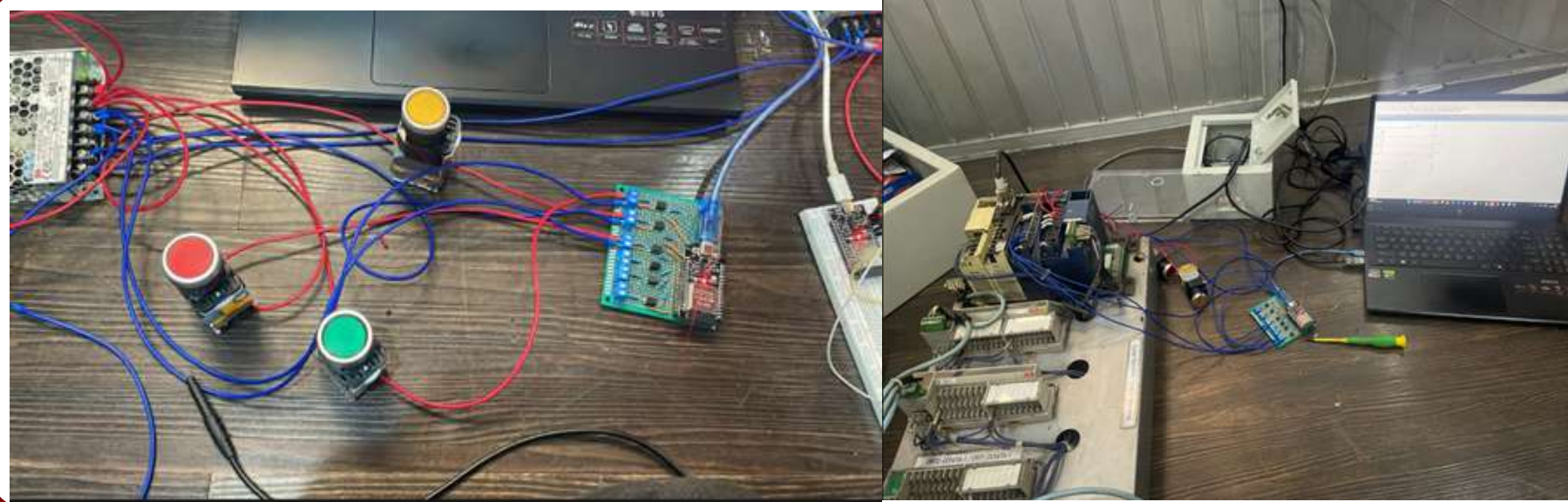
```

OASIS STATUS : auto_line
OASIS STATUS : finish
  
```

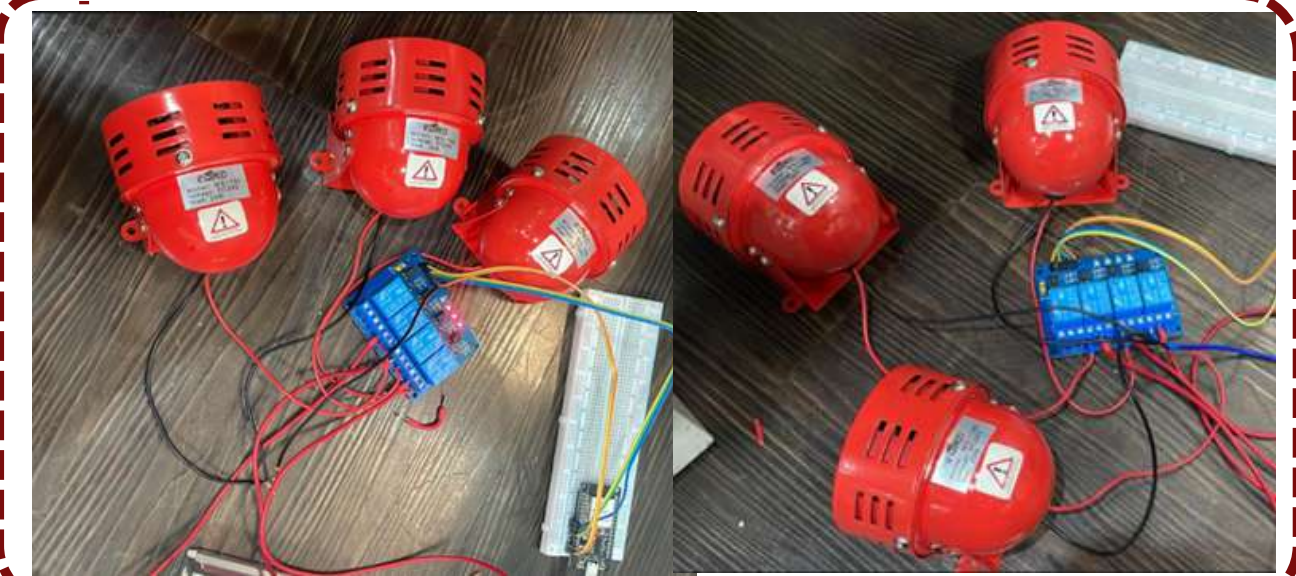


HASIL PERCOBAAN

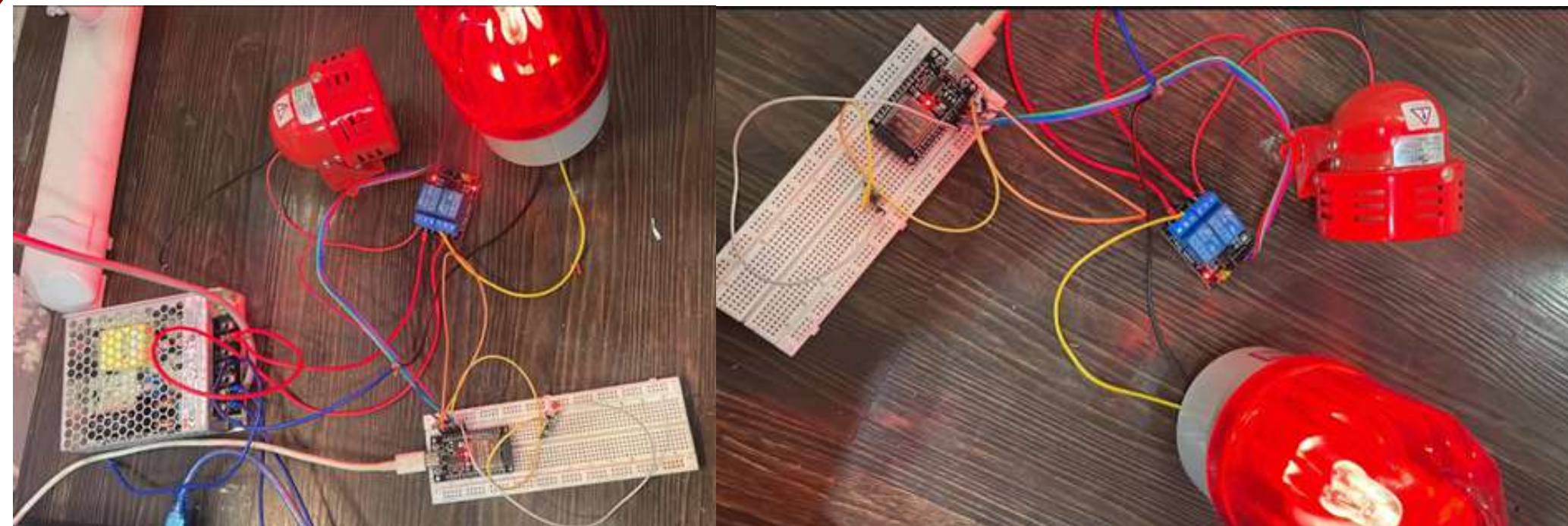
ESP MESIN



ESP LINE

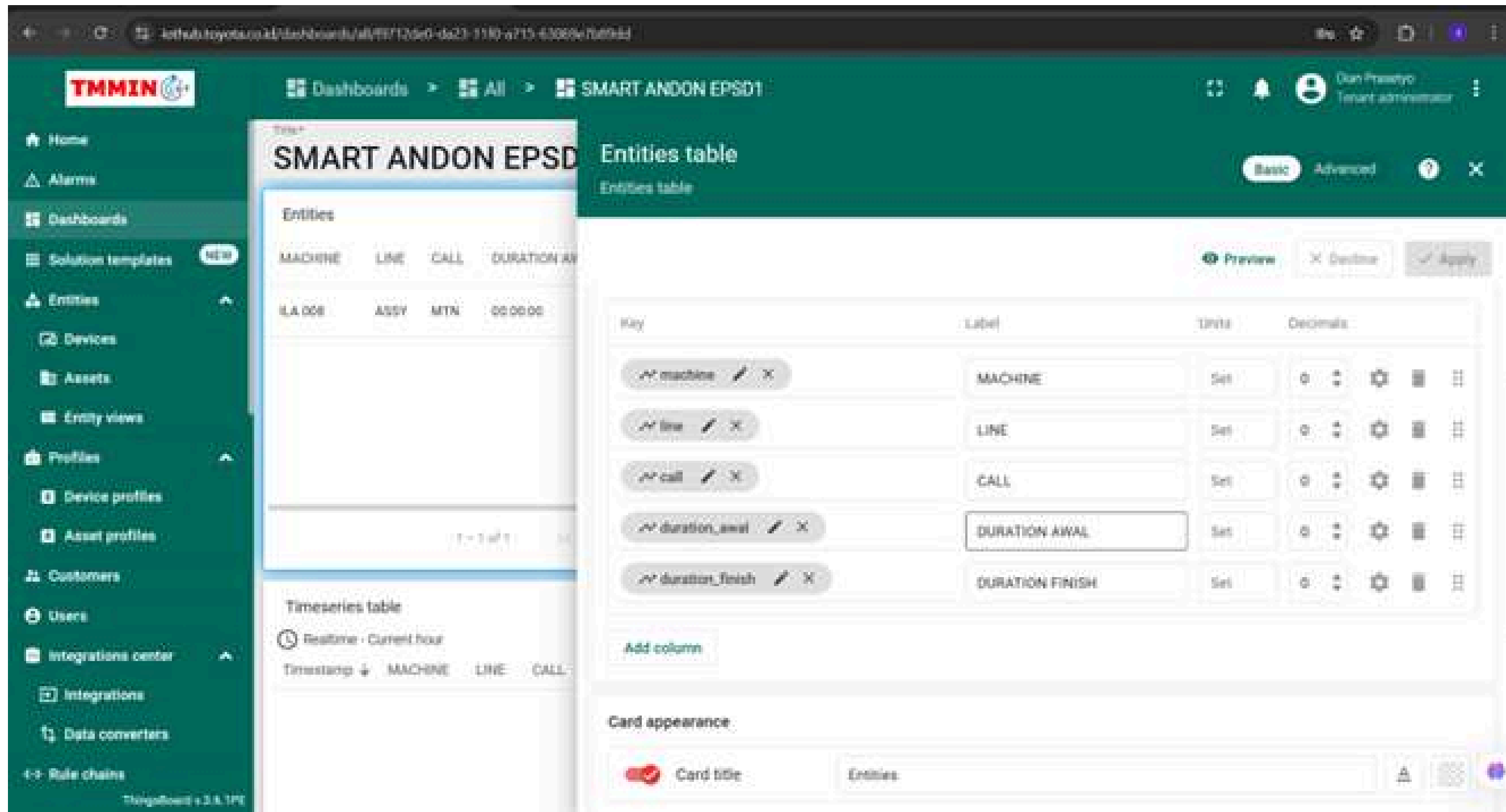


ESP OASIS





DATA YANG DIKIRIM KE THINGSBOARD



TMMIN Dashboards > All > SMART ANDON EPSD1

Entities table

MACHINE	LINE	CALL	DURATION AWAL
ILA 008	ASSY	MTN	00:00:00

Timeseries table

Realtime - Current hour

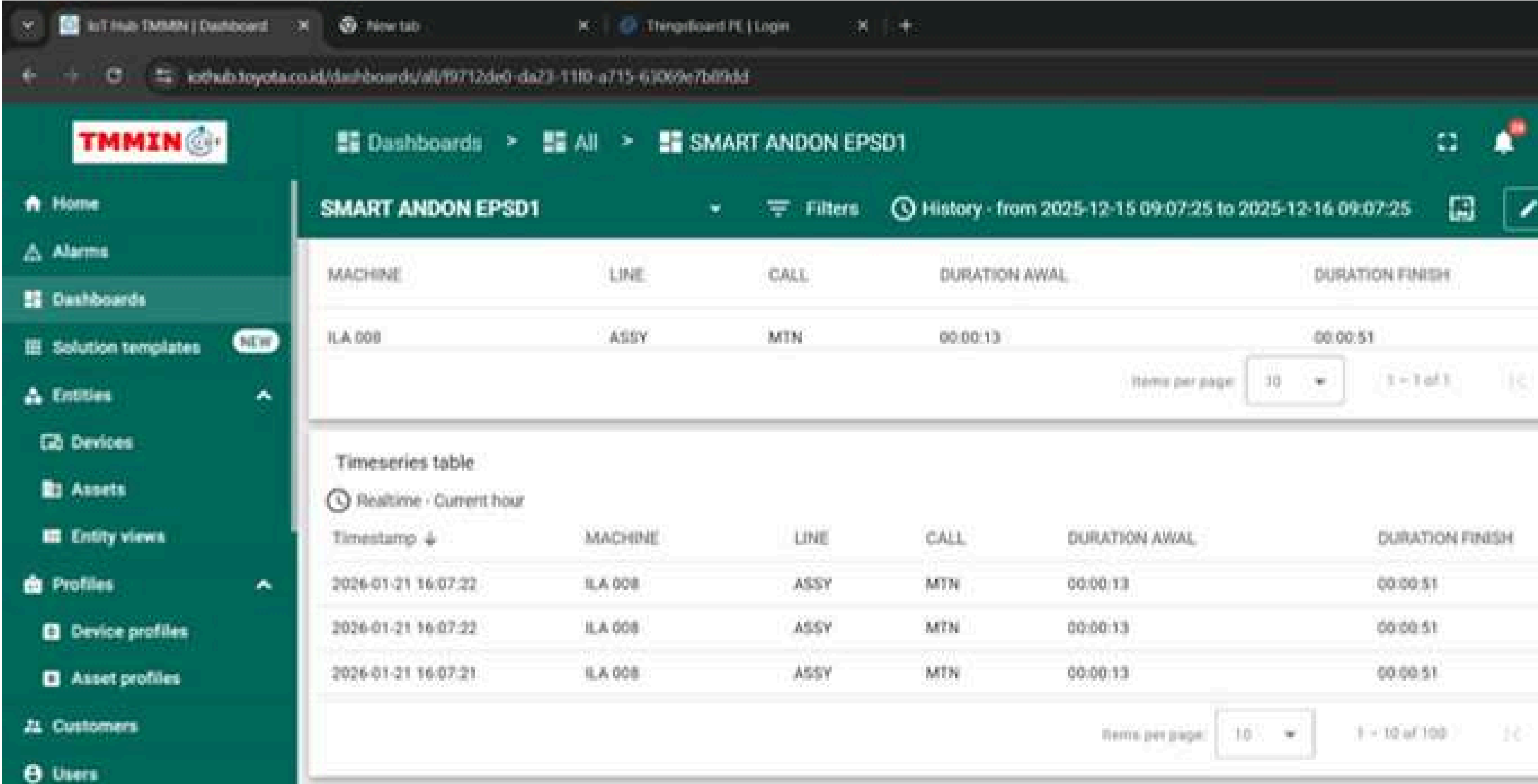
Timestamp	MACHINE	LINE	CALL
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Card appearance

Card title: Entities



TAMPILAN THINGSBOARD



The screenshot displays the TMMIN Thingsboard dashboard. The left sidebar contains navigation links: Home, Alarms, Dashboards, Solution templates (NEW), Entities (Devices, Assets, Entity views), Profiles (Device profiles, Asset profiles), Customers, and Users. The main content area is titled 'SMART ANDON EPSD1' and includes a 'History' filter for the period from 2025-12-15 09:07:25 to 2025-12-16 09:07:25. Below this, there is a table with columns: MACHINE, LINE, CALL, DURATION AWAL, and DURATION FINISH. The table shows one entry for machine ILA 008 on line ASSY with call MTN, showing a duration of 00:00:13 and a finish time of 00:00:51. Below this table is a 'Timeseries table' section with a 'Realtime - Current hour' filter. It shows a table with columns: Timestamp, MACHINE, LINE, CALL, DURATION AWAL, and DURATION FINISH. The table shows three entries for machine ILA 008 on line ASSY with call MTN, showing a duration of 00:00:13 and a finish time of 00:00:51. The bottom right of the dashboard shows pagination: 'Items per page: 10' and '1 - 10 of 100'.

MACHINE	LINE	CALL	DURATION AWAL	DURATION FINISH
ILA 008	ASSY	MTN	00:00:13	00:00:51

Items per page: 10 1 - 1 of 1

Timeseries table

Realtime - Current hour

Timestamp	MACHINE	LINE	CALL	DURATION AWAL	DURATION FINISH
2026-01-21 16:07:22	ILA 008	ASSY	MTN	00:00:13	00:00:51
2026-01-21 16:07:22	ILA 008	ASSY	MTN	00:00:13	00:00:51
2026-01-21 16:07:21	ILA 008	ASSY	MTN	00:00:13	00:00:51

Items per page: 10 1 - 10 of 100



KESIMPULAN

Sistem Smart Andon yang dikembangkan mampu mendeteksi kondisi gangguan mesin secara akurat melalui ESP Mesin, baik yang berasal dari sinyal output PLC maupun dari input pushbutton, dengan menerapkan kesetaraan logika di mana output PLC bernilai logika 1 diperlakukan sebagai kondisi gangguan aktif dan logika 0 sebagai kondisi normal, dengan menggunakan komunikasi MQTT.





THANK YOU

ANY QUESTION?

